

Grammatical markers and inflectional categories: Reconciling the two perspectives

MARTIN HASPELMATH

Max Planck Institute for Evolutionary Anthropology

draft, 2025 April 9

Morphologists often distinguish between two seemingly incompatible ways of viewing word structures: (i) the morph-based view, which typically describes morphological patterns by grammatical markers that are added to roots, and (ii) the paradigm-based view, which typically describes morphological patterns as realizing inflectional categories. This paper notes that these two views need not be incompatible: Everyone agrees that morphs (roots and markers) arise from paradigms, and everyone agrees that inflected forms are not always composed concatenatively of roots and affixes. Markers and categories represent two different ways of talking, but ultimately, all of morphosyntax relies on basic notions such as meanings, forms, classes, constructions, and paradigms. “Categories” are either (classes of) inflectional meanings or classes of inflected forms. The two perspectives can thus be reconciled and need not be seen as being in competition or even mutually incompatible.

1. Two ways of talking about grammar: markers and categories

One of the most basic aspects of grammar is the expression of abstract meanings such as plural reference or past-time reference by “inflectional means”, as illustrated in (1) and (2).

(1) expression of plural reference

English	<i>garden-s</i>	(cf. singular <i>garden</i>)
Latin	<i>hort-i</i>	(cf. singular <i>hort-us</i> ‘garden’)
Arabic	<i>ḥadaaʔiq</i>	(cf. singular <i>ḥadiiqa</i> ‘garden’)

(2) expression of past-time reference

English	<i>you sai-d</i>	(cf. present-time <i>you say</i>)
Turkish	<i>de-di-n</i>	(cf. present-time <i>di-yor-su</i> ‘you say’)
Spanish	<i>dij-iste</i>	(cf. present-time <i>dice-s</i> ‘you say’)

The central observation and starting point of this paper is that linguists talk in two rather different, and seemingly incompatible, ways about such constructions. On the one hand, they talk about them in terms of GRAMMATICAL MARKERS (such as suffixes) and their “combinations”, and they say things like (3a-c).

- (3) a. In most English nouns, plural is expressed by the marker *-s*.
- b. Latin has a nominative plural suffix *-i*.
- c. Past tense is expressed in Turkish by adding the suffix *-di* to the verb root.

On the other hand, they talk about them in terms of INFLECTIONAL CATEGORIES, LEXEMES and the “realization” of their “properties”, and they say things like (4a-c).

- (4) a. English nouns do not inflect for case, but there is a number category with the two properties singular and plural.

- b. In many Arabic noun lexemes, the plural is realized by a specific vowel pattern (“broken plurals”).
- c. In Spanish verb lexemes, the past-tense property is expressed cumulatively with the second person singular property (*-iste*).

Having two different ways of talking about inflectionally expressed meanings has sometimes been perceived as confusing, because the terminology is very uneven (as stressed by Carstairs-McCarthy 2000), and because they are not mutually compatible (as noted by Spencer 2019: §11.2.1): Grammatical markers are kinds of MINIMAL FORMS (or MORPHS, Haspelmath 2020), but inflectional categories (or features) are generally thought of as constituting INFLECTIONAL PARADIGMS with cells that can be filled in diverse ways. When Spanish verbs are presented in paradigms as in (5) (lexeme *decir* ‘say’), there is no immediate need to divide words into morph constituents.

(5) Spanish

	PRESENT TENSE	PAST TENSE	
SG 1	<i>digo</i>	<i>dije</i>	‘I say; I said’ etc.
2	<i>dices</i>	<i>dijiste</i>	
3	<i>dice</i>	<i>dijo</i>	
PL 1	<i>decimos</i>	<i>dijimos</i>	
2	<i>decís</i>	<i>dijistes</i>	
3	<i>dicen</i>	<i>dijeron</i>	

These two different ways of talking about the phenomena are sometimes associated with different and incompatible ways of conceptualizing inflectional regularities, often called “different theories of inflection”. Anderson (2019) puts it as follows:

“Some theorists see the analysis of words as basically a matter of the analysis of minimal meaningful pieces of form – morphemes – ..., and the principles for combining these elements as essentially continuous with the regularities of the syntax. Such “syntax of words” theories would include work by scholars such as Selkirk (1982), Lieber (1980, 1992), and Williams (1981), among others. The alternative on which morphology is described directly as a matter of relations between words not mediated by combination of basic atomic elements like morphemes is typified by the proposals of Zwicky (1985), Anderson (1992), Stump (2001), and others.” (Anderson 2019: §2.4)

In the following, I will call the two views the “morph-based view” (which mostly relies on markers) and the “paradigm-based view” (which mostly relies on categories). The basic distinction has been around in our field for many decades, and can be traced at least to Hockett’s (1954) distinction between an “item and arrangement” and a “word and paradigm” view.¹

Concretely, the morph-based view will lead us to formulate rules such as (6a), where concatenation of morphs is indicated by the plus (+) symbol, and the paradigm-based view will lead us to formulate rules such as (6b), where the category-values [+2] and [+SG] are realized by appending */iste/* to the stem that is represented by the variable *X* (see Anderson 1992: 72 for this notation).

¹ Stewart (2016: 5) distinguishes between “morpheme-based” and “word/lexeme-based” views. Note that the morph-based view is sometimes also called “lexical view” (e.g. Stump 2019), as morphs have often thought to be “stored in the lexicon”. However, the idea of a separate “lexical component” (distinct from grammatical rules or constructions) is questionable, and in approaches such as Jackendoff & Audring (2020) (which they call the “Parallel Architecture”), words, larger fixed expressions and constructions are all treated uniformly (they are all *inventorial items*, in the terms of Haspelmath 2024a).

(6) describing Spanish *dijiste* ‘you said’:

- a. morph-based: verb root + *iste* ‘2SG’
 (concatenation of morphs)
- b. paradigm-based: /X/ [+2, +SG] → /Xiste/
 (realization of category-values)

The goal of the present paper is to point out that these two different views need not be seen as mutually incompatible “theoretical approaches”, but can be treated as different perspectives that can be adopted simultaneously without contradiction. For example, we can treat Spanish *dijiste* ‘you said’ either as a member of the inflectional paradigm in (5), or as consisting of the two morphs *dij-* ‘say (past tense)’ and *-iste* ‘2SG’, or as both at the same time. These perspectives or views or descriptions are both well-motivated and are best treated as complementary rather than as in competition (let alone in conflict) with each other.

In the remainder of this paper, I will make a number of general arguments that concern the foundations of morphosyntax, and especially of the structure of “words”. I will first present some basic concepts of general morphosyntax that are shared by almost all linguists (§2), briefly giving reasons for skepticism about the basicness of “words”, “word-formation rules” and “feature realization”. Then I will make eight points:

- All of morphosyntax is ultimately PARADIGM-BASED (§3);
- inflectional patterns are identified as such by INFLECTIONAL MEANINGS (§4);
- inflectional meanings are typically expressed by INFLECTIONAL MARKERS (§5);
- grammatical markers can be AFFIXES or CLITICS, though only affixes count as inflectional (§6);
- sets of contrasting forms with meanings from the same meaning class constitute INFLECTIONAL PARADIGMS (§7);
- inflectional paradigms can include markerless forms which express inflectional meanings and are thus INFLECTED FORMS as well (§7);
- inflectional meanings expressed in inflectional paradigms may alternatively be called INFLECTIONAL PROPERTIES, and meaning classes may be called INFLECTIONAL CATEGORIES, but the terminology in the field is very inconsistent (§8);
- inflectional meanings can also be expressed in other ways that deviate from the one-to-one canon (§9);
- morphological “operations” cannot be used as comparative concepts (§10).

My concern in this paper is not with different “theories”; instead, I want throw some light on the different ways of talking about inflectional meanings and inflected forms. The key terms that are highlighted in the bullet list above should be compatible with different “theoretical approaches”, so the present paper can perhaps contribute to bridging the apparent gulfs between different views and different research traditions.

From §4 onward, some readers may feel that this paper takes the position of defending the morph-based view. This is not so, as I regard both views as compatible and merely different in terminology and notation, as well as perhaps in some esthetic ways. However, my primary concern is with comparative concepts for general linguistics, not with the description of particular languages. For general-comparative purposes, concepts that involve processes or operations are generally poorly suited, because they presuppose directionality or abstract underlying structures (cf. §10). Moreover, I do not see a good justification for starting out from “words” and for treating morphological patterns as

essentially different from syntactic patterns (see Haspelmath 2011). Especially among linguists who self-identify as “morphologists”, there seems to be a strong tendency to express a clear preference for the paradigm-based view, and I do not see this preference as well-motivated.

To conclude the paper in §11, I will say a few more words about how I see the relation between the two competing perspectives and theoretical goals of grammatical research. Perhaps the most important point is that morphs arise from paradigms, and that we all need both paradigms and morphs for morphosyntax.

2. Seven basic notions of morphosyntax

Most discussions of inflectional patterns start out from the notion of “word” and are couched within “theoretical morphology” (e.g. Anderson 2015; Spencer 2019; Stump 2019), but “word” is not a basic notion of morphosyntax. The criteria for distinguishing between words and non-word expressions (affixes and phrases) are often unclear or different from language to language. As a result, it is questionable whether a rigorous definition of “word” can be of deeper significance for grammatical theorizing (see Haspelmath 2011; 2023b: §5). These doubts about the importance of the “word” also lead me to skepticism about the strong preference for the “word-based/lexeme-based/paradigm-based” view that “has been prevalent among practicing morphologists” (Aronoff 2007: 806).

Instead, I propose that we can reduce most or all notions of morphosyntax to the seven basic notions in Table 1.

meanings
shapes
forms (or expressions)
boundness
classes (of forms or constructions)
constructions
constructional paradigms

Table 1. Seven basic notions of morphosyntax

That forms and constructions are basic is recognized by virtually everyone, and it is also clear that a form is a pairing of a shape and a meaning (for some discussion of the terms *form* and *construction*, see Haspelmath 2020: §4; Haspelmath 2023c). That roots, affixes and clitics are best defined in terms of a distinction between bound forms and free forms was suggested in some of my earlier papers (Haspelmath 2021a; 2023d; 2025b). The notion of “word” can be defined in terms of these more basic notions, but it is not itself taken as a basic notion here.²

Everyone also recognizes classes as basic to morphosyntax (and to language systems in general): classes of shapes, of meanings, and of forms, as well as classes of constructions. Classes of forms that are simultaneously related in terms of meaning and shape (or *form-classes*, in Bloomfield’s 1933 terminology) are also called “categories”. (This term will be discussed in §8 below).

² One may suspect that orthographic words have the function of enhancing readability, but whether a “word” notion can be seen as foundational to morphology (and to the morphology-syntax distinction) is quite unclear (see also Bruening 2018 for related discussion of the “lexicalist hypothesis”). In Audring & Masini’s (2019) handbook of morphological theory, no attempt is made to justify the existence of morphology as a separate domain, and it is simply presupposed.

In addition to forms, classes and constructions, morphosyntax needs paradigms of constructions, as illustrated by the person-inflection paradigm in (7) (from Turkish, cf. the example in (2)) and the English passive paradigm in (8). Both paradigms are simplified and are presented here only for concreteness.

- | | | | |
|-----|-------------------|---------------|--------------------------------|
| (7) | [V- <i>di-m</i>] | ‘I verbed’ | (e.g. <i>dedim</i> ‘I said’) |
| | [V- <i>di-n</i>] | ‘you verbed’ | (e.g. <i>dedin</i> ‘you said’) |
| | [V- <i>di</i>] | ‘s/he verbed’ | (e.g. <i>dedi</i> ‘s/he said’) |
-
- | | | | |
|-----|---|---|--------------------------------------|
| (8) | [NP _α V NP _β] | ‘X _α verbs Y _β ’ | (e.g. <i>They took it.</i>) |
| | [NP _β V-PASS NP _α] | ‘Y _β is verbed by X _α ’ | (e.g. <i>It was taken by them.</i>) |

Constructional paradigms expressing inflectional meanings as in (7) are typically called *inflectional paradigms*, and paradigms that relate different valency patterns as in (8) have sometimes been called *transformations*. The parallelism between inflectional paradigms and syntactic paradigms of this type was long ignored by most linguists, but it has recently come into focus in constructionist approaches (constructional paradigm links have been called “horizontal relations”, Diessel 2019: §10.2; or “relational links” among “sister constructions”, Audring 2019; or relations among “second order schemas”, Booij & Masini 2015).

So we need the seven basic concepts of Table 1,³ but what we may not need in morphosyntax is “words”, “zeroes”, “word-formation rules”, and “realization rules”. In general, we may not need processes of any kind, even though linguists often use process terminology (such as “movement”, “insertion”, “deletion”, “operation”, “derivation”, “conversion”, “readjustment”). Perhaps the biggest problem with process concepts is that they force us to adopt a directionality that is not necessarily present in the language system (see also §10).

Thus, on the present view of morphosyntax there is no basic syntax-morphology distinction, and the traditional distinction between inflection and derivation is not taken as basic either (Haspelmath 2024b). We do need the notion of the *inventorium* (the set of all forms and constructions that cannot be predicted; see n. 1), but this notion is broader than morphosyntax and is thus not listed in Table 1.

On the basis of the seven basic notions of Table 1, I will argue that a view of grammar in terms of markers is not incompatible with a view of grammar in terms of paradigms. In addition and while doing this, I will discuss the following terms (and some of the problems with some of them):

- | | | |
|-----|---|---|
| (9) | a. inflectional meanings (§4) | (e.g. plural meaning) |
| | b. inflectional markers (§5) | (e.g. a dative case marker) |
| | c. clitic and affix markers (§6) | (e.g. prepositions and case affixes) |
| | d. inflectional paradigms (§7) | (e.g. the paradigms of Russian <i>kniga</i> ‘book’) |
| | d. inflected forms (§7) | (e.g. a past tense form) |
| | f. inflectional properties (§8.1) | (e.g. the plural (category-)property) |
| | g. inflectional categories (§8.1) | (e.g. the number (meaning-)category) |
| | h. inflectional category-classes (§8.2) | (e.g. the Spanish Future tense) |
| | i. morphological operations (§10) | (e.g. a rule of affixation) |

³ In addition, it seems that the notion of *productivity* is crucial, though it has a different status than the notions in Table 1, because it is not an all-or-nothing matter.

We will see that most of these terms have their justification and can be defined in a clear way that is easy to understand, on the basis of the basic notions in Table 1. As the terms are not used uniformly across the discipline, it is unavoidable that the definitions that I give here do not always correspond to the actual use in the literature, but I think that they are as close as is possible (given the goal of providing clear definitions).

Let us now move on, in the next section, to the most fundamental aspect of all of morphosyntax, the existence of paradigmatic relations with meaning-shape parallels.

3. Paradigmatic relations as the foundation of morphosyntax: Morphs arise from paradigms

While the morph-based view and the paradigm-based view have often been contrasted in the morphological literature (§1), there can be no doubt that paradigmatic relations (in the broad sense) are the foundation of all of morphosyntax. Segmenting word-forms into morphs is impossible without paradigmatic sets of related word-forms. This can be illustrated by the English word *unbelievable*, which Mithun (2014: 72) uses to introduce the notion of a morph:⁴

- (10) *unbelievable*: *believ-able* ‘able to be believed’
 un-believ-able ‘not able to be believed’

These segmentations are straightforward and uncontroversial, but it is worth pointing out that they are based on paradigmatic word sets such as those in (11a) and (11b). If we did not have multiple pairs of words of this kind, there would be no basis for the breaking up *unbelievable* into morphs as in (10).

- (11)a. *believe* *believ-able* ‘able to be believed’
 wash *wash-able* ‘able to be washed’
 count *count-able* ‘able to be counted’
- b. *believable* *un-believable* ‘not believable’
 happy *un-happy* ‘not happy’
 aware *un-aware* ‘not aware’

The basis of morphological structure in paradigmatic relations was noted by de Saussure (1916: 178-179), who pointed out that the French word *dé-faire* ‘undo’ would become unanalyzable if all other words containing *dé-* ‘un-’ and *faire* ‘do’ disappeared from the language. Morphs disappear when paradigms disappear.

In exactly the same way, paradigmatic relations are the foundation of syntactic segmentation and constituency. Consider the sentences in (12).

- (12) a. *Flying planes can be dangerous.* (‘planes that fly’)
 Falling leaves can look beautiful.
 Smoking passengers can be fined.
- b. *Flying planes can be dangerous.* (‘to fly planes’)
 Analyzing sentences can be difficult.
 Helping children can be necessary.

⁴ Mithun uses the term *morpheme*, but what she means is a morph (i.e. a minimal form), so I use the latter term here. (I avoid “morpheme”, because it is used in different senses.)

If English did not have paradigmatic sets such as those in (12a) and (12b), there would be no basis for setting up the abstract constructional patterns [V-*ing* N]_{NP} ‘N that Vs’ and [V-*ing* NP]_{VP} ‘to V NP’, and we would not be able to relate the well-known ambiguity of the sentence *Flying planes can be dangerous* to different syntactic structures.

In the older literature, paradigmatic relations were often described in terms of the notion of “analogy”. This is particularly clear in Hermann Paul’s (1880) book, which locates the foundation of morphological and syntactic structure in PROPORTIONAL GROUPINGS such as those in (13) (see Auer & Murray 2015: 194).

- (13) *cat* : *cats*
rat : *rats*
cap : *caps*

A similar notation (with a colon between related forms that share a root) is used by Lyons (1968: 182), who discusses proportions such as the following:

- (14) *tall* : *taller* : *tallest*
big : *bigger* : *biggest*
nice : *nicer* : *nicest*
bad : *worse* : *worst*

Linguists do not always make it explicit that morphological and syntactic segmentation is based on such paradigmatic (or analogical relations), but the basic point is not controversial. While a *morpheme* is sometimes said to be “a minimal unit of meaning” (e.g. Yule 2010: 290), we would not set up morphemes if we did not have paradigms with covariation of both meanings and shapes as in (11)-(14). For example, we can set up purely semantic paradigms as in (15) (from Fillmore 2006: §5.2), but these do not give rise to segmentation.

- (15) *horse* : *stallion* : *mare*
sheep : *ram* : *ewe*
cattle : *bull* : *cow*
swine : *sow* : *boar*

Thus, segmentation (and thus morphology and syntax) is based on the recurrent observation that “some linguistic forms bear partial phonetic-semantic resemblances to other forms” (Bloomfield 1933: 160). The relations between these partially similar forms are paradigmatic relations, but the result is that we arrive at constituents, and finally at the minimal constituents: the morphs.⁵ We can express this as a slogan, which can be taken as summarizing the key point of this paper:

- (16) Morphs arise from paradigms.

However, what happens with paradigms such as (14), which seem to include one opposition that is more like (15), with no phonetic resemblance, namely *bad* : *worse*? It is non-straightforward paradigmatic relations such as these that have given rise to

⁵ In his pioneering paper, Bloomfield (1926: 155) defined a morpheme as “a minimum form”, or “a recurrent (meaningful) form which cannot in turn be analyzed into smaller recurrent (meaningful) forms”. This is the same as the definition of *morph* in Haspelmath (2020) (Bloomfield did not use the term *morph*, but his “morpheme” was basically the same as today’s morph).

different ways of talking about morphosyntactic patterns, and that will occupy us during much of this paper.

4. Inflectional meanings

Before embarking on a longer discussion of inflectional markers and paradigms, let us delimit the range of meanings that are considered inflectional here. Eight semantic domains are listed in (17), with a few examples of meanings belonging to each of the domains.

(17) a. case (or role) meanings	e.g. nominative, accusative, ergative, dative, genitive, instrumental
b. person meanings	e.g. first, second, third
c. number meanings	e.g. singular, plural, dual
d. gender meanings	e.g. masculine, feminine, neuter, animate, inanimate
e. tense meanings	e.g. past, present, future
f. mood meanings	e.g. indicative, imperative, interrogative
g. polarity meanings	negative, affirmative
h. evidentiality meanings	e.g. visual, hearsay, inferred

Within each semantic domain, the meanings are contrastive, i.e. they cannot be present simultaneously. But inflected forms may of course express several meanings from different domains simultaneously, e.g. Turkish *de-di-n* ‘you (SG) said’ in (2), which expresses past (from the tense domain), second (from the person domain) and singular (from the number domain).

The delimitation of inflectional meanings takes the form of a list because a content-based delimitation of inflectional and derivational meanings seems to be the only possibility. Linguists have often presupposed that inflection and derivation are two different types (or even modules) of morphology,⁶ but there is no clear conceptual basis for this (Haspelmath 2024b). Thus, for current purposes, it seems best to delimit the range of phenomena that are considered inflectional in a semantic and semi-arbitrary way. What matters here is that the meanings in (17) are uncontroversially among the meanings that are most commonly considered inflectional.⁷ When other meanings are expressed by affixes (including all those that are usually considered “derivational”), they are usually treated as markers, not in terms of categories or category-properties, so they are not particularly relevant to the topic of this paper.⁸

⁶ For example, Aikhenvald (2015: 51) says that “a category or a process may be inflectional in one language and derivational in another”, but there are no uniformly applicable nonsemantic criteria for this distinction.

⁷ In Indo-European languages, voice constructions (active, passive, middle) are often considered inflectional as well, but in most other languages, they pattern with derivational constructions such as causative and applicative constructions. Aspect meanings (e.g. habitual, progressive, perfective) are often considered inflectional in languages around the world, so they could be added to the list in (17). But they are often linked with tense meanings in an inextricable way (as in the Spanish perfective past *dijimos*, which contrasts with an imperfective past *decíamos*), and some aspect meanings are usually considered derivational (e.g. semelfactive). They are left out here for simplicity because the list in (17) is big enough. Finally, some transpositional patterns (e.g. participles and nominalizations) are widely considered inflectional, but there are also many that are considered derivational.

⁸ Derivational affixes are not normally called “derivational markers” or “grammatical markers”, even though they are usually considered to be part of the grammatical system of a language. It seems that this is

The meanings in (17) are defined independently of morphological patterns, so some of them can also be expressed by free forms. For example, person meanings can be expressed by independent personal pronouns (*me, you, us*, etc.), number meanings can be expressed by numerals or other quantifiers (e.g. *two, several*), and temporal meanings can be expressed by adverbials (*yesterday, in the future*). However, some of them can be said to be more properly syntactic than semantic: Some case forms do not seem to have a clear semantic interpretation (e.g. when a language requires a genitive case form in combination with an adposition), and gender markers are not clearly semantic either, though gender classes always have a semantic core. Thus, it would be an oversimplification to say that the inflectional meanings are purely semantic. They convey what has sometimes been called *syntacticosemantic content* (e.g. Stump 2016: 28). When inflectional meanings are expressed by inflected forms (in inflectional paradigms), they are also called *grammemes* (e.g. Mel'čuk 2006: 23; Malchukov 2011; Plungian 2011: 10), or *inflectional properties*, or even “inflectional categories”. The various terms that have been used for them in the literature will be discussed in §8 below.

It should be noted that inflectional meanings may also be expressed by periphrastic constructions, as in the French future-tense construction *je vais danser* ‘I will dance’. Such constructions have sometimes posed conceptual problems, and they will be discussed further in the next section.

5. Inflectional markers

For inflectional markers, I propose the definition in (18), which builds on the notion of inflectional meaning that we saw in the preceding section.

(18) inflectional marker

An inflectional marker is an affix that expresses an inflectional meaning.

Affixes are minimal bound forms that are not roots and that do not occur on different kinds of roots (Haspelmath 2021a). Thus, the markers that we saw in (1) and (2) above (English *-s*, Latin *-i*, Turkish *-di* etc.) are inflectional markers by this definition.

In addition, grammatical markers can also be auxiliary elements which are not affixes, e.g. French *vais* in the periphrastic future-tense construction *je vais danser* ‘I will dance’, but such elements are not considered inflectional markers, and they do not generally constitute elements of an inflectional paradigm. This is in line with a long tradition that does not, for example, say that Spanish has an inflectional paradigm of “preposed case markers” as in (19).

- (19) Spanish
- | | |
|---------------------|--------------|
| <i>la mujer</i> | (nominative) |
| <i>a la mujer</i> | (dative) |
| <i>de la mujer</i> | (genitive) |
| <i>con la mujer</i> | (comitative) |

because they are generally thought of as “creating new words” rather than as expressing grammatical meanings. (And as Stump (2019: §4.6) observes, derivational patterns are not normally treated in terms of “feature realization”, though again there do not seem to be any deep reasons for this.) Thus, everything that I say in this paper could in principle be applied to derivational marking as well, and it seems to be an accident of history that linguists rarely talk about “derivational categories” and “realization of derivational properties”.

The set of forms in (19) can be regarded as a “syntactic paradigm” of nominal (and prepositional) phrases, because the prepositions are words rather than prefixes (they typically precede nouns, but they may also precede adjectives, as in *la pobre mujer* ‘the poor woman’). They express inflectional meanings, but they do not constitute an inflectional paradigm.

Auxiliary verbs and adpositions are grammatical markers, but they are not inflectional markers, as they are clitics, not affixes.⁹ We can easily talk about these elements in the “marker mode”, and we do not talk about them frequently in the “category mode”. For example, we do not often say that the French future-tense construction with *vais* is an expression or “realization” of a future tense property or category. And we almost never say that Spanish prepositions as in (19) express or “realize” case properties. (There is no deep reason for this, and these ways of talking may actually have become more common in recent decades.)

6. Grammatical marking beyond markers: Clitics, affixes, endophony and zero

Inflectional paradigms have traditionally been at the core of grammatical descriptions of languages such as Latin and Arabic, but as we just saw, meanings such as ‘plural’, ‘past time’, ‘accusative’, ‘interrogative’ and ‘negative’ can also be expressed by grammatical markers that are not affixes but clitics. (20) shows a few further examples.

(20)	a. dative preposition	Spanish	<i>a la mujer</i>	‘to the woman (DAT)’
	b. plural word	Haitian Creole	<i>liv yo</i>	‘books’
	c. past-tense proclitic	Seychelles Creole	<i>nou ti fini</i>	‘we finished’
	d. interrogative clitic	Polish	<i>czy wiesz?</i>	‘do you know?’
	e. negative particle	German	<i>komm nicht</i>	‘don’t come’

Such grammatical markers behave like inflectional affixes in many ways, and especially in typological linguistics, the two types of grammatical markers (affixes and clitics) are often treated together (for examples, case affixes and adpositions are treated together as *flags*, and negative affixes and particles are treated together as *negators*). A term such as *syncretism* that used to be restricted to inflection is now routinely used also for other grammatical markers (e.g. Baunaz & Lander 2018)

When morphologists deal with inflectional paradigms (e.g. Blevins 2015), they limit themselves to paradigms of words (inflected forms), but in grammatical descriptions of the world’s languages, linguists often present paradigms of clitics as well. For example, Philippova (2021: 144) presents a verbal paradigm of Amri Karbi with four person proclitics in the first slot, as seen in (21), and Meyerhoff (2013) gives a verbal tense-aspect-mood paradigm of Bislama, as seen in (22).

(21) Amri Karbi (Philippova 2021: 144)

<i>nang</i> =	2nd
<i>ne</i> =	1st
<i>i</i> =	1st plural inclusive
<i>hali</i> =	1st plural exclusive

⁹ See Haspelmath (2023d) for a definition of *clitic*, and Haspelmath (2026) for a definition of *auxiliary*.

- (22) Bislama (Meyerhoff 2013)
- | | |
|-----------------|---------------------|
| <i>(bam)bae</i> | irrealis |
| <i>bin</i> | anterior |
| <i>finis</i> | completive |
| <i>save</i> | habitual, abilitive |
| <i>stap</i> | imperfective |
| <i>Ø</i> | non-progressive |

The inflectional meaning types that we saw in (17) above (case/role, person, number, gender, tense, mood, polarity, evidentiality) are thus just as relevant to non-affixal marking as to affixal marking. Moreover, just as inflectional paradigms can include cases of suppletion, this may happen with paradigms involving clitic markers. For example, while English shows a suppletive comparative form of ‘bad’ that contrasts with a regular affixed form with *-er*, Italian shows a suppletive comparative form of ‘good’ that contrasts with a regular form with proclitic *più*.

- (23) English
- | | | |
|-------------|--|---------------------|
| <i>warm</i> | | <i>warm-er</i> |
| <i>bad</i> | | <i>worse</i> |
- (24) Italian
- | | | | |
|--------------|--------|------------------------|----------|
| <i>caldo</i> | ‘warm’ | <i>più caldo</i> | ‘warmer’ |
| <i>buono</i> | ‘good’ | <i>migliore</i> | ‘better’ |

Similarly, while Lezgian shows a suppletive negative form of ‘be’ that contrasts with a regular affixed form with *-č*, Korean shows a suppletive negative form of ‘know’ that contrasts with a regular form with the proclitic word *ani*.

- (25) Lezgian
- | | | | |
|--------------|-----------|-------------------|-------------------|
| <i>žezwa</i> | ‘becomes’ | <i>žezwa-č</i> | ‘does not become’ |
| <i>ja</i> | ‘is’ | <i>tuš</i> | ‘is not’ |
- (26) Korean (Chung 2009)
- | | | | |
|------------|--------|---------------------|------------|
| <i>ka-</i> | ‘go’ | <i>ani ka-</i> | ‘not go’ |
| <i>al-</i> | ‘know’ | <i>moll-</i> | ‘not know’ |

While such cases are not commonly discussed in general treatments of grammatical paradigms, they are very common in the world’s languages, and it does not seem to be helpful to exclude them from consideration when trying to understand the coding of inflectional meanings. A purely “word-based” perspective is thus insufficient when dealing with inflectional meanings.

That some inflectional paradigms include periphrastic forms is well-known and has sometimes been discussed (e.g. Haspelmath 2000; Chumakina & Corbett (eds.) 2013). An example comes from Bulgarian, which is similar to Modern Greek (see (28) below) except that the future marker *šte* is a proclitic, unlike the Greek prefix *tha-*.

- (27) Bulgarian (Popova & Spencer 2013: 196)
- | | | |
|--------------------|---------------------|----------------|
| <i>mislj-a</i> | [think-1SG.PRS] | ‘I think’ |
| <i>misl-ix</i> | [think-1SG.AOR] | ‘I thought’ |
| <i>šte mislj-a</i> | [FUT think-1SG.PRS] | ‘I will think’ |

Thus, category terms such as *case/role*, *number*, *tense*, or *mood*, which were traditionally restricted to inflectional paradigms, are generally used more broadly, for other kinds of grammatical marking. In addition to tense inflection, languages may have non-inflectional tense marking; in addition to number inflection, languages may have non-inflectional number marking, and so on.

Marking is thus a term that covers both clitic marking and inflectional marking. The latter covers both affixal marking and nonconcatenative marking, as we will see in the next section.

7. Inflectional paradigms and inflected forms

In addition to affixes, languages have NONCONCATENATIVE ways of expressing inflectional meanings, in particular zero-coding (illustrated in 28a-b), endophonic coding (illustrated in 29a-b), and reduplicative coding (illustrated in 30a-b). When an inflectional meaning is expressed in one of these ways while related meanings are expressed by inflectional markers, linguists often include them as full members of INFLECTIONAL PARADIGMS, and we can call them INFLECTED FORMS. All these instantiate what has been called NONCONCATENATIVE INFLECTION (e.g. Bickel & Nichols 2007: 181-182). These kinds of patterns have been an important argument against a morph-based view of inflection, and in favour of the paradigm-based view. However, these forms are compatible with both views, though we need the additional notion of an *inflected form*.

Zero-coding is illustrated by a tense paradigm from Modern Greek and a case paradigm from Turkish.

(28) a. Modern Greek

<i>Ø-yráf-o</i>	[PRS-write-1SG]	‘I write’ (present tense)
<i>é-yráf-a</i>	[PST-write-1SG.PST]	‘I wrote’ (past tense)
<i>tha-yráf-o</i>	[FUT-write-1SG]	‘I will write’ (future tense)

b. Turkish

<i>ev-Ø</i>	[house-NOM]	‘(the) house’ (nominative)
<i>ev-in</i>	[house-GEN]	‘of (the) house’
<i>ev-de</i>	[house-LOC]	‘in (the) house’

Endophonic coding is illustrated by a number paradigm from German (showing vocalic endophony, or metaphony) and a person paradigm from Chatino (showing tonal endophony; the superscript digits indicate different tones).

(29) a. German

<i>Garten</i>	[garden.SG]	‘garden’
<i>Gärten</i>	[garden.PL]	‘gardens’

b. Chatino (of San Juan Quiahije; Cruz & Stump 2018: 211)

<i>skon</i> ⁴⁰	[arm.1SG]	‘my arm’
<i>skon</i> ¹	[arm.2SG]	‘your arm’
<i>skon</i> ²	[arm.3SG]	‘his/her arm’

Reduplicative coding is illustrated by a tense paradigm from Latin and a number paradigm from Agta.

- (30) a. Latin
 curr-o [run.PRS-1SG] ‘I run’
 cu~curr-i [PRF~run-1SG] ‘I ran’
- b. Agta (Philippinic; Healy 1960: 7)
 uffu [thigh] ‘thigh’
 uf~uffu [PL~thigh] ‘thighs’

Thus, as inflectional paradigms contain not only forms characterized by an inflectional marker, we need a concept of INFLECTED FORM that is broader. A definition is given in (31).¹⁰

(31) **inflected form**

An inflected form is a combination of a root and affixes (or reduplicants), or just a root, which is associated with specific inflectional meanings and which contrasts with a range of other forms in which inflectional meanings of the same domain are expressed by affixes.

A zero-coded form like English *house-Ø* is an inflected form because it has a specific inflectional meaning: plural number. Similarly, the prefixless Greek form *Ø-yráf-o* [PRS-write-1SG] can be said to be zero-coded for tense because it has a specific inflectional meaning: present tense. Zero-coding is possible here because the contrasting meanings (singular and plural number; future tense and past tense) are obligatorily expressed by affixes, which means that the affixless form can have a specific meaning (cf. Bybee 1994).¹¹

Similarly, a vowel-alternating pair such as German *Garten/Gärten* ‘garden(s)’ in (29a) is said to constitute two inflected forms because there are other pairs where the same inflectional meanings are expressed by affixes, e.g. *Tag/Tag-e* ‘day(s)’, *Straße/Straße-n* ‘street(s)’. The same applies to tone-alternating pairs and reduplicative pairs.

By contrast, if an inflectional meaning is never expressed by affixes, the forms are not inflected forms according to the definition in (31). For example, Uduk (a Koman language of South Sudan) has a handful of nouns with different singular and plural forms (e.g. *mì/mìyà* ‘goat(s)’, *wàthi?/kwāni* ‘man/men’, *à ’bóm/ūp* ‘woman/women’; Killian 2015: 63), but there is no recurring affix, so these do not constitute an inflectional paradigm (a set of inflected forms). Similarly, Nynorsk Norwegian has different subject and object forms of some personal pronouns (as seen in 32), but there are no affixes, so these do not count as inflected forms.

¹⁰ This definition only covers inflected forms based on a root like those in (28)–(30). One should ultimately also include inflected forms based on composite lexeme-stems, e.g. English *real-iz-ed*, *bak-er-s*. This can be done by replacing *root* in (31) by *lexeme-stem* (a notion that is defined in Haspelmath 2024b: 68).

¹¹ If a form that lacks a marker is not opposed to an obligatory contrasting form with an affix, it thus cannot be said to be an inflected form. For example, the Japanese plural suffix *-tati* is optional (e.g. Nakanishi & Tomioka 2004), so when a noun does not bear this suffix, it cannot be said to have a specific (inflectional) meaning, and it is thus not an inflected form (and there is no zero-coding). This is why the text used the English example *house-Ø*, not the Turkish example *ev* in (28b); Turkish seems to be more like Japanese, so that the representation *ev-Ø* [house-NOM] is problematic. Similarly, when a language has a case affix but there is no specific meaning associated with affixless forms, the affixless form cannot be said to be an inflected form. For example, Persian has an accusative marker *-râ*, but nominals lacking this marker are not associated with a specific meaning (Arkadiev 2023), so we cannot say that they exhibit zero coding.

(32) Nynorsk Norwegian subject and object forms of personal pronouns

	subject	object
'I'	<i>eg</i>	<i>meg</i>
'you'	<i>du</i>	<i>deg</i>
'he'	<i>han</i>	<i>han</i>
'she'	<i>ho</i>	<i>ho</i>
'we'	<i>vi</i>	<i>oss</i>
'you'	<i>de</i>	<i>dykk</i>
'they'	<i>dei</i>	<i>dei</i>

Thus, in the conceptual system proposed here, inflectional markers are foundational for inflectional paradigms. This is in line with the usual practice of linguists: We do not set up morphological paradigms when there are no markers. For example, we do not say that there is a derivational relationship in kinship paradigms such as (33). The meaning relationships are uniform (male vs. female) in the four pairs, but there is no corresponding formal relationship and thus no morphological pattern, even though one could say that there is a “derivational meaning” involved (also recall the similar pattern in (15) above).

(33) English kinship terms

<i>father</i>	<i>mother</i>
<i>son</i>	<i>daughter</i>
<i>brother</i>	<i>sister</i>
<i>uncle</i>	<i>aunt</i>

The situation is the same with inflection: We identify inflectional relationships on the basis of inflectional affixes in paradigms, not on the basis of inflectional meanings alone.¹² In this sense, inflectional affixes (or markers) have priority over other ways of expressing inflectional meanings.¹³

The notion of an *inflected form* is thus more abstract than the concrete notions *inflectional meaning* (§4) and *inflectional marker/affix* (§5). It is based on the general practice of linguists, so it should not be controversial. But it does not look like a primitive notion, and many linguists have preferred other abstract notions, such as the idea of an abstract “morpheme” or abstract “feature values” that need to be “realized”. Such alternative ways of talking about inflectional patterns may well have advantages, but the point of this paper is that there is no need to dispense with the concept of an inflectional marker. And as we saw, inflectional paradigms (virtually) always contain some markers, so that markers are in a sense more basic than nonconcatenative marking.

8. Inflectional (category-)properties and (meaning-)categories

Let us now go back to the problematic term “category” that is used in the title of this paper. “Categories” have also been called “features”, and the “paradigm-based” view of morphology has typically advocated a “feature realization” view, instead of a view where

¹² There may be situations where inflectional meanings are expressed only by endophony or by reduplication. Especially situations where only tone marks an inflectional meaning are probably not so uncommon. This might eventually lead to a more complex formulation of (31), but I leave this complication aside here.

¹³ An interesting question is whether there can also be “syntactic meanings” that are zero-expressed. This question has often been answered affirmatively (e.g. by Mel’čuk 1979, but also by many generative linguists), but I will not pursue it here.

inflectional patterns are based on minimal forms (*markers*, i.e. bound morphs that are not roots). But what exactly are “features”, or “categories”?¹⁴ Confusingly, the term “category” has been used in at least three different senses that are relevant for morphosyntax: For inflectional meanings (or properties), for meaning domains (or dimensions), and for language-particular form-classes. This section seeks to clarify these different uses of “category”.

Inflected forms are typically described in tabular exemplary paradigms as in (34), which are familiar from pedagogical grammars. There is no controversy around them, but linguists do not quite know what general label to use for inflectional meanings (such as *preterite*) and meaning domains (such as *tense*).

(34) Polish tense inflection (some inflected forms of *robić* ‘do’)

		PRESENT	PRETERITE
SINGULAR	1ST	<i>robi-ę</i>	<i>robi-la-m</i>
	2ND	<i>robi-sz</i>	<i>robi-la-ś</i>
	3RD	<i>robi</i>	<i>robi-la</i>
PLURAL	1ST	<i>robi-my</i>	<i>robi-ły-śmy</i>
	2ND	<i>robi-cie</i>	<i>robi-ły-ście</i>
	3RD	<i>robi-q</i>	<i>robi-ły</i>

8.1. Inflectional meanings and meaning domains

We first look at inflectional paradigms from a general-comparative perspective, before moving to a language-particular perspective in §8.2. For the general perspective, we need to start from inflectional meanings such as ‘present’, ‘preterite’, ‘singular’, ‘plural’, and ‘1st’, ‘2nd’, ‘3rd’. These are sometimes called INFLECTIONAL PROPERTIES, and meaning domains such as ‘tense’ (e.g. ‘present’, ‘preterite’), number (‘singular’, ‘plural’), and person (‘1st’, ‘2nd’, ‘3rd’) are often called INFLECTIONAL CATEGORIES (or FEATURES). The term pair *property/category* is perhaps the most readily recognizable pair, but different linguists have used different terms. In Table 2 (partially following Carstairs-McCarthy 2000: 267), I give a number of different term pairs. It can be seen that “category” has been used both for inflectional meanings and for inflectional meaning domains, so one needs to be very careful with this term.¹⁵ For clarity, one can perhaps use the compound terms *category-property* (for meanings such as ‘preterite’ or ‘plural’) and *meaning-category* (for inflectional domains such as tense or number).

inflectional meaning = (<i>category</i> -)property	inflectional domain = (<i>meaning</i> -)category	(representative authors)
property	category	Matthews (1972; 1991)
grammeme	category	Mel’čuk (2006)
category	category complex	Wurzel (1984; 1989)
grammatical meaning	grammatical category	Bybee (1986)
value	feature	Zwicky (1985), Corbett (2012)
category	dimension	Haspelmath (2002)
morphosyntactic property	inflectional category	Stump (2019)

Table 2. Terms for category-properties and meaning-categories

¹⁴ It seems that the notion of a “morphological feature” was not current before the 1980s (Anderson 1982 and Zwicky 1985 seem to have been influential in spreading it). Matthews (1972; 1991) exclusively used “category” and “property”.

¹⁵ Van der Auwera & Gast (2011) suggest that the two uses of *category* can be distinguished by calling the first “basic categories” (e.g. singular, dative), and the second “higher categories” (e.g. number, case).

While tabular inflectional paradigms as in (32) have long been used in grammatical descriptions, the term *category* is much more recent. Until the 19th century, it was still common to call the domains of inflectional meanings *accidents*, following the grammatical description of Donatus in the 5th century (Lyons 1968: 198; Matthews 1991: 190). It appears that the term *category* for inflectional meanings or meaning domains was not common before Jespersen and Bloomfield (van der Auwera & Gast 2011: 169).

I suspect that some confusion arose because the term *category* can be used in science in two different ways:

- (35) category
 - a. general notion
 - b. class (= a set of entities that have something in common)

Originally, a category was only a broad notional concept, such as Aristotle's and Kant's substance, quantity, quality, place, action and so on (Thomasson 2022). It appears that this older notional sense (not the form-class sense) underlies the idea that categories are meanings or meaning domains. If they are meanings, they can be said to be *marked* or *expressed* by certain forms or constructions. For example, linguists often say things such as the following (emphasis added):

- (36) a. "The more often two categories are attested being *expressed* by one morpheme in the world's languages, the closer these two categories will end up in the splits graph." (Cysouw 2011: 439)
- b. "A grammatical category may be *marked* just by tone alternation. In Yuhup, perfective and imperfective aspect forms differ just in their tone: rising for the former, falling for the latter." (Aikhenvald 2015: 74)
- c. "Chadic languages do not *mark* the category of evidentiality." (Frajzyngier 2016: 260)
- d. "Just like other categories such as definiteness or aspect, irrealis is not systematically *expressed* by grammatical means in all languages." (von Prince et al. 2022: 230)

For greater clarity, one could replace *category* in such uses by *meaning* or *meaning domain*. But as I noted, *category* is also used in the sense of a form-class (see §8.2), so perhaps when linguists say that "category X is expressed/marked" in a certain way, they mean that the form-class is characterized in a certain way, rather than that a meaning is expressed in some way.

The terms VALUE and FEATURE (Zwicky 1985; Corbett 2012) are usually clearer than "category" (though "feature" is sometimes also used in the sense of "feature value"), but they are more abstract, so that it is not immediately transparent that the values generally refer to mutually exclusive contrastive INFLECTIONAL MEANINGS. And in fact, the term *feature* has also been used for distinguishing different inflectional classes, and such class features have sometimes been treated together with semantic features (e.g. Müller 2004; Corbett 2015). In this tradition, the expression of inflectional meanings (and possibly other, more abstract values) is often called REALIZATION, and alternatively realization rules are often called EXPONENCE rules.¹⁶ What exactly is covered by the notions of exponence and by the features and their values is often unclear in this tradition because

¹⁶ It seems that this way of talking about inflection goes back to Zwicky (1985). Zwicky says, for example: "The following principle of English is a typical rule of exponence: In the context of [CAT:verb]. [VFORM:past] is realized by the suffixation of /d/" (1985: 372). (Anderson 1982 did not talk about "realization" or "exponence", but used other terms such as "developing".)

of the high degree of abstractness of the chosen features.¹⁷ In the present context, abstract information is not so relevant because general and cross-linguistic research on morphosyntactic patterns must be based on concrete comparative concepts. (This is not to deny that for the purposes of language-particular analysis, abstract notions such as zeroes, defaults and operations may often be useful.)

8.2. Language-particular classes of inflected forms (“grams”)

In addition to meanings or meaning domains, “categories” can refer to FORM-CLASSES, i.e. classes of morphs, words, phrases or clauses.¹⁸ Bloomfield (1933: 270) originally defined *category* simply as a subtype of form-class:

“Large form-classes which completely subdivide either the whole lexicon or some important form-class into form-classes of approximately equal size, are called categories... Thus, ... singular and plural substantives [are categories], since these two form-classes, of approximately equal size, completely subdivide the form-class of substantives.”

In the Bloomfieldian tradition, *category* is also used for classes of lexemes (lexical categories, e.g. Baker & Croft 2017) or for classes of phrases (syntactic categories, e.g. Gazdar 1981). But for inflectional morphology, the relevant sense is that of a class of inflected forms.

Examples of language-particular form-classes of the relevant type are given in (37). The forms in these classes are characterized by rather different markers, but they all express the same inflectional meanings.

- (37) some Latin form-classes
- a. dative forms: *homin-i* ‘to the person’, *lup-o* ‘to the wolf’, *reg-ibus* ‘to the kings’
 - b. future forms: *lauda-b-is* ‘you will praise’, *audi-e-s* ‘you will hear’
 - c. plural forms: *lup-i* ‘wolves’, *homin-es* ‘persons’, *mens-ae* ‘tables’
 - d. 1st person forms: *laud-o* ‘I praise’, *lauda-mus* ‘we praise’, *audi-o* ‘I hear’

We can say that “the Latin dative” is a category (in the sense of a form-class) that includes all the inflected forms that express dative meaning, or that “the Latin plural” is a category that includes all the forms that express plural meaning.

If a category is a set of forms rather than a meaning (or meaning domain), we can say that it expresses a meaning, e.g. that “the German dative expresses the recipient”, or that “the French subjunctive expresses notions of doubt or irrealis”. Language-particular form-classes of the “category” type can perhaps be called “category-classes”, contrasting with the “meaning-categories” that we saw in §8.1. While category-classes can be used to express meanings, meaning-categories are conversely expressed by forms. There is thus a striking difference between the two senses, but this difference is rarely discussed by linguists.¹⁹

One approach in which the difference was highlighted is the one adopted by Bybee & Dahl (1989), Bybee et al. (1994) and Dahl (2000), as well as some other work in this

¹⁷ Trommer (2012: 1) describes *exponence* as the way in which “abstract morphosyntactic information is realized by phonological structure”.

¹⁸ The term *form-class* is not often used these days, but it was briefly mentioned in §2 above.

¹⁹ Note also that in this sense, category-class terms can be used in peculiar locative-like constructions such as “Latin *lupo* stands in the dative”, or “in the dative, Latin *lupus* has an *-o* ending” (see Haspelmath 2024b: 47 for some discussion). Moreover, we even say things like “she uses the future tense a lot” (cf. Dahl 2016: 429), meaning that “she uses many future-tense forms”.

tradition. Bybee & Dahl introduced the term GRAM for a language-particular form-class sharing an inflectional meaning and observe:

“Many terms used in grammar may be understood as ambiguous between referring to 'notional' ('semantic') or 'grammatical' categories. Thus 'progressive' may either denote a certain meaning or context of use (which presumably can occur in any language) or a category, like the English Progressive, which has a certain meaning and a certain expression. Our 'grams' are entities of the second kind.” (Bybee & Dahl 1989: 51-52).

Thus, *gram* can be used as an alternative term for category-classes. Bybee & Dahl presented it as an abbreviation of “grammatical morpheme”, though Dahl (1985: 21) had used “grammatical category” (see also Dahl 2000: 23, n. 2). In order to compare grams across languages, these authors use a notion of “gram type” (e.g. Dahl 2022).

Confusingly, linguists also sometimes talk about “higher” inflectional categories such as tense and modality as *expressing* certain notions, as in the following passages (emphasis added):

- (38) a. “For Langacker, the question whether the category of tense *expresses* either modal or temporal semantics is not really an either/or issue.” (Fortuin & Boogaart 2016: 522)
- b. “Participant-internal possibility modality therefore *expresses* an ability or capability of some event participant.” (Gluckman & Bowler 2020)
- c. “One approach to the relationship between the two domains is that evidentiality is a single linguistic category which *expresses* concepts from both domains. This is evidentiality in the broad sense...” (Brugmann & Macaulay 2015)

In such contexts, the term *category* seems to refer to the entire collection of relevant form-classes (i.e. grams) from different languages that express inflectional meanings from the relevant domain.

We have seen in this section that the terminology surrounding inflectional paradigms is diverse and often confusing, and it may be best to simply avoid the term *category* in the context of grammatical marking (even though I used it in the title of this paper). Elements such as ‘plural’, ‘accusative’ and ‘imperative’ are perhaps best called INFLECTIONAL MEANINGS, and the higher domains are perhaps best called INFLECTIONAL DIMENSIONS. If the term “category” is used, it may be useful to distinguish between MEANING-CATEGORIES, CATEGORY-PROPERTIES (§8.1), and CATEGORY-CLASSES (§8.2).

It appears that part of the reason why the terminology is so diverse and confusing is that linguists are still very much influenced by the tradition of Latin linguistics, where inflectional patterns (declensions and conjugations) were at the core of systematic school teaching. Latin has a very peculiar way of expressing inflectional meanings, and the terminology used for Latin does not generalize easily to languages that have rather different structures.

9. Deviations from one-to-one mapping of shapes and meanings

In discussions of the “word-based” approach to inflection, it has often been observed that inflectional meanings and inflectional markers are not always in a one-to-one relationship. This has been taken as an argument for a “realizational” approach, where inflectional properties are not expressed by inflectional morphs but by morphological processes or operations (as seen in (6b) above; e.g. Anderson 2015; Stump 2019). In

particular, zero expression, cumulative expression, and syncretism have been taken as arguments for a view according to which the meanings of complex words cannot be derived from their parts as they are in syntax.²⁰

However, morphological processes come with their own problems (discussed in the next section), and deviations from one-to-one relationships are not specific to “word structure”, as they occur in syntactic patterns as well. This section will primarily discuss cumulative expression (§9.1), syncretism (§9.2), and inflectional classes (§9.3), but I will also say a few words about multiple exponence and a few other patterns (§9.4).

9.1. Cumulative expression

Let us first look at a few examples of cumulative expression (or SYNEXPRESSION, Haspelmath 2023a) in some partial inflectional paradigms:

(39) a. cumulative expression of person meanings and number meanings

Russian

1+SG	<i>piš-u</i>	‘I write’
2+SG	<i>piš-eš’</i>	‘you (SG) write’
1+PL	<i>piš-em</i>	‘we write’
2+PL	<i>piš-ete</i>	‘you (PL) write’

b. cumulative expression of case meanings and number meanings

Latin

NOM+SG	<i>hort-us</i>	‘(the) garden’
DAT+SG	<i>hort-o</i>	‘to (the) garden’
NOM+PL	<i>hort-i</i>	‘(the) gardens’
DAT+PL	<i>hort-is</i>	‘to (the) gardens’

c. cumulative expression of tense meanings and polarity meanings

Coptic

PRF+AFF	<i>a-f-sôtm</i>	‘he heard’
PRF+NEG	<i>mp-f-sôtm</i>	‘he didn’t hear’
HAB+AFF	<i>ša-f-sôtm</i>	‘he used to hear’
HAB+NEG	<i>me-f-sôtm</i>	‘he did not used to hear’

Such patterns are familiar, but cumulative expression is also found outside of inflectional paradigms. The most famous case of cumulative expression of meanings that are otherwise expressed by non-affix forms is the behaviour of French prepositions *de* and *à*, which combine regularly with the feminine singular article, but which do not occur with masculine and plural articles. Instead, French has cumulative “preposition-article complexes”:

²⁰ For example, Anderson (2004) says that “If morphological structure were as represented by the morpheme-based picture, we would also expect to find that the relation between components of form and those of content would be one-to-one.” He therefore favours a realizational view, where morphology is less syntax-like: “Adopting such a theoretical position reduces considerably the apparent similarity between the structures of words and of sentences, and accordingly leads to a much less “syntactic” picture of morphology.”

(40) cumulative expression of spatial meaning and definiteness

French

to.the(F.SG)	<i>à la</i>	
to.the(M.SG)	<i>au</i> [o]	(e.g. <i>au revoir</i> lit. ‘to the re-seeing’)
to.the(PL)	<i>aux</i> [o]	
from.the(F.SG)	<i>de la</i>	
from.the(M.SG)	<i>du</i> [dy]	
from.the(PL)	<i>des</i> [de]	

This kind situation may not be very common, but the example shows that cumulative expression is not restricted to “word structure”. Clearly, inflectional affixes must be allowed to express multiple inflectional meanings simultaneously, but this is a more general phenomenon that extends beyond inflectional paradigms. (Recall also the examples of suppletion in (23)-(26) above, which show that suppletion extends beyond inflectional paradigms.)

9.2. Syncretic expression

Another deviation from one-to-one mapping is syncretism (or COEXPRESSION, Haspelmath 2023a), as illustrated by the German person-number paradigms in (41) (illustrating present-tense forms of ‘come’ and ‘give’) and the Arabic case paradigms in (42) (*al-baytu* ‘the house’, *al-bayt-aani* ‘the two houses’).

(41) German

1SG	<i>komm-e</i>	<i>geb-e</i>
2SG	<i>komm-st</i>	<i>gib-st</i>
3SG	<i>komm-t</i>	<i>gib-t</i>
1PL	<i>komm-en</i>	<i>geb-en</i>
2PL	<i>komm-t</i>	<i>geb-t</i>
3PL	<i>komm-en</i>	<i>geb-en</i>

(42) Standard Arabic (Ryding 2005: 184; 188)

NOM	<i>al-bayt-u</i>	<i>al-bayt-aani</i>
ACC	<i>al-bayt-a</i>	<i>al-bayt-ayni</i>
GEN	<i>al-bayt-i</i>	<i>al-bayt-ayni</i>

In the German paradigm, the elements *-t* and *-en* have two different person-number functions, and in Arabic, *-ayni* is used for accusative and genitive.

The simplest view of such situations is that they show HOMONYMY of inflectional markers, e.g. that German two different suffixes *-t* (3rd person singular *-t* and 2nd person plural *-t*), and that Arabic has two different suffixes *-a* (accusative suffix *-a* and genitive suffix *-a*). Many linguists do not find such descriptions satisfactory, but they are always possible: Morph-based descriptions of syncretic patterns may be less “insightful” than paradigm-based descriptions that use additional devices (such as underspecification), but whether an analysis is insightful or not is often a matter of esthetic judgement.

Linguists typically prefer descriptions in terms of underspecification, which are sometimes straightforward, e.g. when the German suffix *-n* is described as [+plural], while 2nd plural *-t* is described as [+2nd, +plural]. In this case, the suffix *-n* is the simple elsewhere case, and nothing more needs to be said. But in many situations, other questions arise, e.g. which category-properties (or feature values) should be assumed to begin with.

This can lead to many complications (see, e.g., Corbett 2012), and it is unclear whether the issues are specific to morphology. The distinction between inflectional homonymy and inflectional underspecification is essentially the same distinction that is described elsewhere as the distinction between homonymy/polysemy and vagueness/indeterminacy (e.g. Zhang 1998; Vicente & Falkum 2017; among many others), and the issues are very much the same as in lexemic semantics. Thus, whether any morphology-specific concepts or mechanisms are needed is unclear.

9.3. Affix suppletion (requiring inflectional classes)

Inflectional meanings are often expressed differently depending on the class of lexemes on which they occur, giving rise to inflectional classes. For example, Cree has two different plural suffixes (-*wak* and -*a*), as shown in (39); and Amele has about 30 different classes of nouns that determine different forms of adpossessionive person affixes (‘my wife’, ‘your flesh’, ‘his groin’, etc.):

(43) Cree (Algonquian; Okimāsis 2018: 19-20)

SG	PL	
<i>atim</i>	<i>atim-wak</i>	‘dog(s)’
<i>aikāk</i>	<i>aikāk-wak</i>	‘skunk(s)’
<i>astotin</i>	<i>astotin-a</i>	‘hat(s)’
<i>misit</i>	<i>misit-a</i>	‘foot/feet’

(44) Amele (Roberts 1987: 172-175)

1SG	<i>aide-ni</i>	<i>cuhu-ni</i>	<i>gailo-mi</i>	<i>cot-i</i>
2SG	<i>aide-n</i>	<i>cuhu-nin</i>	<i>gailo-m</i>	<i>cot-in</i>
3SG	<i>aide-g</i>	<i>cuhu-n</i>	<i>gailo-h</i>	<i>cot-ig</i>
	‘wife’	‘flesh’	‘groin’	‘brother’

Inflectional classes are well-known from Latin and other Indo-European languages, and they used to be known as “declension classes” and “conjugation classes”. They take up a lot of the effort of beginning learners, but again, they do not exhibit morphology-specific issues. They are similar to suppletion and other kinds of synonymic expression, and they can be dealt with in a morph-based perspective or in a paradigm-based perspective.

9.4. Other deviations from the canonical ideal

Some other kinds of deviation from one-to-one expression of meanings are listed in (45) (see, e.g., Anderson 1992: §3.2; Hippisley & Stump 2016: §1).

- (45) a. empty morphs (e.g. theme vowels such as -*i*- in Polish *rob-i-sz*)
 b. multiple exponence (e.g. English *child-r-en*)
 c. circumfixation (e.g. German *ge-schrieb-en* ‘written’)
 d. dependency (e.g. Latin *hort-or* ‘I encourage’ not: ‘I am encouraged’)

“Empty morphs” are a problem only if one assumes that all material in a larger form must be part of a morph, but this assumption is not made here. Multiple exponence is a problem only if one assumes that the occurrence of one marker blocks the occurrence of another marker, but “blocking” is not assumed here. Circumfixation can be seen as the

cooccurrence of a prefix and a suffix in a single (partially idiomatic) construction. Deponency can be treated as homonymy (as in §9.2).

Thus, special additional mechanisms are not required, and these phenomena do not distinguish morphology from syntax (or words from phrases) anyway. “Empty markers” and “multiple marking” are common in syntax as well. Thus, they cannot be seen as strong reasons for a special word-based (or paradigm-based) approach that is unique to word structures.

10. Morphological “operations”

Since Hockett (1954), linguists have been aware that there is a tension between process-based characterizations of morphosyntactic patterns and static characterizations. Hockett noted that some authors use descriptions such as those in (46a), while others use descriptions as in (46b) (the example is inspired by Haas 1946; see Hockett 1954: 211; 213)).²¹

- (46) a. The morphological processes in Tunica include juxtaposition, prefixation, infixation, and suffixation.
b. Tunica morphemes fall into several position classes: stems, prefixes, infixes, and suffixes.

It would seem that these two “ways of speaking” are not more than notational variants, though later authors have often given deeper significance to morphological “operations” such as “deletion”, “truncation”, “readjustment”, or even “morphological movement”. In the generative approach following Chomsky & Halle (1968), rule sequencing has often played an important role. For example, in Distributed Morphology, it is crucial that the operation of “vocabulary insertion” happens after other morphological operations such as “impoverishment” (e.g. Siddiqi 2019: 150).

In the current context, the problem with morphological “operations” is that these concepts cannot be applied to all languages using the same criteria. Such “operations” may well be motivated for language-particular description, but for general-comparative linguistics, we need concepts that can be applied uniformly in all languages. Concatenative operations such as “prefixation” or “juxtaposition” are not a problem, because they can be translated easily into static parlance using the basic notions of §2. But nonconcatenative concepts such as “stem change” cannot be used as comparative concepts. For example, while it is common to talk about English verb ablaut in process terms as in (47a), the static formulation in (47b) is the only possibility in a general-comparative context.

- (47) a. In the English verb lexeme *SING*, past tense is signaled by a vowel change (*sing* > *sang*).
b. The English verb lexeme *SING* has a present-tense form *sing* and an endophonically related past-tense form *sang*.

As Hockett (1954: 211) noted, process-based characterizations must assign some kind of priority to one of the two forms, and such decisions may have to be made in an arbitrary way. Thus, morphological “operations” are generally incompatible with the idea that comparative concepts must be formulated in a uniform way across languages.

²¹ Tunica is a language of the Mississippi Valley that was described by Mary Haas.

“Realizational” approaches such as Distributed Morphology (Siddiqi 2019) or Paradigm Function Morphology (Stump 2019) may well be suitable for describing or analyzing particular languages in an elegant way, but they do not offer suitable concepts for general-comparative purposes.

11. Concluding remarks: General concepts, theories, and analyses

In this paper, I have discussed inflectional patterns both from the perspective of grammatical markers (more specifically, inflectional affixes) and from the perspective of inflectional paradigms and the meanings that constitute them. I pointed out that all grammatical structure arises from relations in form paradigms whose members exhibit “phonetic-semantic resemblances” (Bloomfield 1933), and that these also give rise to constituents and ultimately to morphs (i.e. minimal forms) (§3).

Constructional patterns in which inflectional meanings (§4) are expressed are called *inflectional paradigms* when the meanings are expressed by affixes (inflectional markers, §5). Such paradigms may also include forms which lack an affix and where the relevant meaning is expressed nonconcatenatively by zero, by endophony or by reduplication (§6). *Inflected forms* need not contain an inflectional affix, but they must occur in a paradigm where some forms contain an affix (§7). In §8, the different senses of the term “category” are highlighted, and in §9, it is shown that different non-concatenative patterns can fit into the sparse ontology of Table 1 (in §2).

Many linguists have taken the view that the nonconcatenative expression of inflectional meanings requires different mechanisms of description. According to Blevins et al. (2019), the gulf between the morph-based and the paradigm-based view is wide:

“While both examine the ‘structure’ of words, they do so from very different perspectives, assuming very different objects of analysis and arriving at very different results.” (Blevins et al. 2019: 265)

But how significant are the differences between the two types of analyses or descriptions? Are they perhaps merely notational variants, so that the choice between them is a matter of taste? This can be answered only by considering the larger goals of the enterprise of general morphosyntax. There can be different goals, especially:

- (i) understanding the nature of morphosyntactic patterns in the world’s languages;
- (ii) providing a formal framework for describing languages elegantly;
- (iii) determining the elements of the innate universal grammar (UG) that makes language acquisition possible

Morphologists do not always say clearly what their general goals are. Goal (iii) (determining the elements of UG for morphosyntax) has been widely adopted, if only implicitly, by generative linguists, but it does not seem to be within easy reach. We do not even know if any grammatical elements are innate, and we do not have a good way of inferring the supposed innate elements from the language structures that we describe.

Goal (ii) seems to be what underlies much work in the tradition of HPSG or LFG and related formalist approaches (e.g. Stump 2001; Spencer 2019), but as there are often many different ways in which analyses can be considered “elegant”, it is not clear how the goal can be stated objectively.

Goal (i), finally, is the goal of general-comparative linguistics in the tradition of Greenberg (1963), and this is what I happen to be particularly interested in (see

Haspelmath 2021b). But for this goal, the type of notation is not really important, and all that matters is that we have comparative concepts that can be applied in the same way in all languages. This is true for the basic notions and the other concepts defined in this paper, but not for the “operations” that I discussed in §10. Thus, one of the reasons for discussing the conceptual relationships between “markers” and “categories” is to have clear comparative concepts that are independent of particular frameworks or research traditions, and that can be used for framework-free typology and theorizing.

But the primary reason for writing this paper is to help reconcile the seeming gulf between morph-based and paradigm-based approaches, which has been presupposed by many researchers in morphology.²² This contrast does not relate clearly to any of the three goals (i)-(iii). It is not needed for formulating universal generalizations of the Greenbergian type, and it does not seem to be highly relevant to proposals about the innate UG (though Anderson 1992 probably assumed innateness). Perhaps most importantly for the present context, there does not seem to be any deep reason for saying that a paradigm-based view (or an approach in terms of feature realization) offers more elegant descriptive possibilities than a view of morphosyntax where morphs play an important role.

Morphologists have often said that morphology is very different from syntax, in that it “realizational”, while syntax is concatenative, e.g.

“Realizational views of morphology are thus quite different from the quasi-syntactic picture of word formation to which we are often tempted. Adopting such a theoretical position, in particular, reduces considerably the apparent similarity between the structures of words and of sentences, and accordingly leads to a much less “syntactic” picture of morphology.” (Anderson 2004: §1.1)

However, we saw that non-concatenative phenomena such as cumulative expression may also affect syntactic patterns (§6), and that only a few additional notions (such as “inflected form”, §7) are needed to describe inflectional patterns. It is quite doubtful that there are principled differences between morphology and syntax.

Proponents of the paradigm-based view may react to the present paper by saying that it confirms their view because it because constructional *paradigms* are a central basic notion here (see Table 1 in §2). However, the notion of an inflectional *marker* is just as basic for describing inflected forms and inflectional patterns, and it is surely not practical to describe all of morphosyntax in terms of realizational “operations”. Thus, I would like to claim that the present proposal offers a good way of reconciling the two seemingly incompatible perspectives: It is not the case that morphology contrasts with syntax in that it must be either morph-based or paradigm-based (or either marker-based or category-based). Instead, all of morphosyntax is in a sense both paradigm-based and morph-based, because morphs (and larger segmental forms) arise from paradigms (see (16) above).

References

- Aikhenvald, Alexandra Y. 2015. *The art of grammar: A practical guide*. Oxford: Oxford University Press.
 Anderson, Stephen R. 1982. Where’s morphology? *Linguistic Inquiry* 13(4). 571–612.
 Anderson, Stephen R. 1992. *A-morphous morphology*. Cambridge: Cambridge University Press.

²² E.g. Aronoff (2007: 803): “Linguists have followed two ways in the study of words. One seeks to accommodate the word, the other to obliterate it.” As this paper makes clear, I do not see the two views as incompatible.

- Anderson, Stephen R. 2004. Towards a less “syntactic” morphology and a more “morphological” syntax. In van Sterkenburg, Piet (ed.), *Linguistics today: Facing a greater challenge*, 31–45. Amsterdam: Benjamins.
- Anderson, Stephen R. 2015. The morpheme: Its nature and use. In Baerman, Matthew (ed.), *The Oxford handbook of inflection*, 11–33. Oxford: Oxford University Press.
- Anderson, Stephen R. 2019. A short history of morphological theory. In Audring, Jenny & Masini, Francesca (eds.), *The Oxford handbook of morphological theory*, 19–33. Oxford: Oxford University Press. (<https://doi.org/10.1093/oxfordhb/9780199668984.001.0001>)
- Aronoff, Mark. 2007. In the beginning was the word. *Language* 83(4). 803–830.
- Audring, Jenny & Masini, Francesca (eds.). 2019. Theory and theories in morphology. *The Oxford handbook of morphological theory*, 1–16. Oxford: Oxford University Press. (<https://doi.org/10.1093/oxfordhb/9780199668984.001.0001>)
- Audring, Jenny. 2019. Mothers or sisters? The encoding of morphological knowledge. *Word Structure*. Edinburgh University Press 12(3). 274–296. (doi:10.3366/word.2019.0150)
- Auer, Peter & Murray, Robert W. (eds.). 2015. *Hermann Paul’s “Principles of language history” revisited: Translations and reflections*. Berlin: De Gruyter. (doi:10.1515/9783110348842)
- Baker, Mark C. & Croft, William. 2017. Lexical categories: Legacy, lacuna, and opportunity for functionalists and formalists. *Annual Review of Linguistics* 3. 179–197.
- Baunaz, Lena & Lander, Eric. 2018. Syncretisms with the nominal complementizer. *Studia Linguistica* 72(3). 537–570. (doi:10.1111/stul.12076)
- Bickel, Balthasar & Nichols, Johanna. 2007. Inflectional morphology. In Shopen, Timothy (ed.), *Language typology and syntactic description*, vol. III: Grammatical categories and the lexicon, 169–240. Cambridge: Cambridge University Press.
- Blevins, James & Ackerman, Farrell & Malouf, Robert. 2019. Word and paradigm morphology. In Audring, Jenny & Masini, Francesca (eds.), *The Oxford handbook of morphological theory*, 265–284. Oxford: Oxford University Press. (<https://doi.org/10.1093/oxfordhb/9780199668984.001.0001>)
- Blevins, James P. 2015. Inflectional paradigms. In Baerman, Matthew (ed.), *The Oxford handbook of inflection*, 87–111. Oxford: Oxford University Press.
- Bloomfield, Leonard. 1926. A set of postulates for the science of language. *Language* 2(3). 153–164. (doi:10.2307/408741)
- Bloomfield, Leonard. 1933. *Language*. New York: H. Holt and Company.
- Booij, Geert & Masini, Francesca. 2015. The role of second order schemas in the construction of complex words. In Bauer, Laurie & Körtvélyessy, Livia & Štekauer, Pavol (eds.), *Semantics of complex words*, 47–66. Cham: Springer.
- Bruening, Benjamin. 2018. The lexicalist hypothesis: Both wrong and superfluous. *Language* 94(1). 1–42. (doi:10.1353/lan.2018.0000)
- Brugman, Claudia M. & Macaulay, Monica. 2015. Characterizing evidentiality. *Linguistic Typology* 19(2). 201–237. (doi:10.1515/lingty-2015-0007)
- Bybee, Joan L. & Dahl, Östen. 1989. The creation of tense and aspect systems in the languages of the world. *Studies in Language* 13(1). 51–103.
- Bybee, Joan L. & Perkins, Revere & Pagliuca, William. 1994. *The evolution of grammar: Tense, aspect and modality in the languages of the world*. Chicago: The University of Chicago Press.
- Bybee, Joan L. 1986. On the nature of grammatical categories: A diachronic perspective. In Choi, Soonja & Devitt, Dan & Janis, Wynn & McCoy, Terry & Zhang, Zheng-sheng (eds.), *Proceedings of the second Eastern States Conference on Linguistics*, 17–34. Buffalo: University at Buffalo.
- Bybee, Joan L. 1994. The grammaticization of zero: Asymmetries in tense and aspect systems. In Pagliuca, William (ed.), *Perspectives on grammaticalization*, vol. 109, 235–254. Amsterdam: Benjamins.
- Carstairs-McCarthy, Andrew. 2000. Category and feature. In Booij, Geert E. & Lehmann, Christian & Mugdan, Joachim (eds.), *Morphologie/Morphology (Volume 1)*, 264–272. Berlin: De Gruyter. (doi:10.1515/9783110111286.1.4.264)

- Chomsky, Noam & Halle, Morris. 1968. *The sound pattern of English*. New York: Harper & Row.
- Chumakina, Marina & Corbett, Greville (eds.). 2013. *Periphrasis: The role of syntax and morphology in paradigms*. Oxford: British Academy.
(doi:10.5871/bacad/9780197265253.001.0001)
- Chung, Inkie. 2009. Suppletive verbal morphology in Korean and the mechanism of vocabulary insertion. *Journal of Linguistics* 45(3). 533–567. (doi:10.1017/S0022226709990028)
- Corbett, Greville G. 2012. *Features*. Cambridge: Cambridge University Press.
- Corbett, Greville G. 2015. Features. In Baerman, Matthew (ed.), *The Oxford handbook of inflection*, 35–46. Oxford: Oxford University Press.
- Cruz, Hilaria & Stump, Gregory. 2018. The morphology of essence predicates in Chatino. In Bonami, Olivier & Boyé, Gilles & Dal, Georgette & Giraudo, Hélène & Namer, Fiammetta (eds.), *The lexeme in descriptive and theoretical morphology*, 203–234. Berlin: Language Science Press. (<https://zenodo.org/records/1407003>)
- Cysouw, Michael. 2011. Understanding transition probabilities. *Linguistic Typology* 15(2). 415–431.
- Dahl, Östen (ed.). 2000. *Tense and aspect in the languages of Europe* (Empirical Approaches to Language Typology: EUROTyp 20–6). Berlin: Mouton de Gruyter.
- Dahl, Östen. 2022. Perfects across languages. *Annual Review of Linguistics* 8(Volume 8, 2022). 279–297. (doi:10.1146/annurev-linguistics-031120-123428)
- de Saussure, Ferdinand. 1916. *Cours de linguistique générale* (ed. par Charles Bally, Albert Sechehaye, Albert Riedinger). Lausanne & Paris: Payot.
- Diessel, Holger. 2019. *The grammar network*. Cambridge: Cambridge University Press.
- Dryer, Matthew S. 2006. Functionalism and the theory-metalanguage confusion. In Wiebe, Grace & Libben, Gary & Priestly, Tom & Smyth, Ron & Wang, Sam (eds.), *Phonology, morphology, and the empirical imperative: Papers in honour of Bruce Derwing*, 27–59. Taipei: The Crane Publishing Company.
(<http://www.acsu.buffalo.edu/~dryer/foundation.htm>)
- Fillmore, Charles J. 2006. Frame semantics. In Geeraerts, Dirk (ed.), *Cognitive Linguistics: Basic readings*, 373–400. Berlin: Mouton de Gruyter.
- Fortuin, Egbert & Boogaart, Ronny. 2016. Modality and Mood in Cognitive Linguistics and Construction Grammars. In Nuyts, Jan & Auwera, Johan van der (eds.), *The Oxford Handbook of Modality and Mood*. Oxford University Press UK.
- Gazdar, Gerald. 1981. On syntactic categories. *Philosophical Transactions of the Royal Society of London (Series B, Biological Sciences)* 295(1077). 267–283.
(doi:doi.org/10.1098/rstb.1981.0139)
- Gluckman, John & Bowler, Margit. 2020. The expression of modality in Logoori. *Journal of African Languages and Linguistics* 41(2). 195–238. (doi:10.1515/jall-2020-2010)
- Greenberg, Joseph H. 1963. Some universals of grammar with particular reference to the order of meaningful elements. In Greenberg, Joseph H. (ed.), *Universals of language*, 73–113. Cambridge, MA: MIT Press.
- Haas, Mary. 1946. A grammatical sketch of Tunica. In Osgood, Cornelius (ed.), *Linguistic structures of Native America*, 337–366. New York: Viking Fund.
- Haspelmath, Martin. 2000. Periphrasis. In Booij, Geert E. & Lehmann, Christian & Mugdan, Joachim (eds.), *Morphology: An international handbook on inflection and word-formation (Volume 1)*, 654–664. Berlin: Walter de Gruyter.
(doi:10.1515/9783110111286.1.9.654)
- Haspelmath, Martin. 2002. *Understanding morphology*. London: Arnold.
(<https://zenodo.org/record/1236482>)
- Haspelmath, Martin. 2011. The indeterminacy of word segmentation and the nature of morphology and syntax. *Folia Linguistica* 45(1). 31–80. (doi:10.1515/flin-2017-1005)
- Haspelmath, Martin. 2020. The morph as a minimal linguistic form. *Morphology* 30(2). 117–134. (doi:10.1007/s11525-020-09355-5)
- Haspelmath, Martin. 2021. Bound forms, welded forms, and affixes: Basic concepts for morphological comparison. *Voprosy Jazykoznanija* 2021(1). 7–28. (doi:10.31857/0373-658X.2021.1.7-28)

- Haspelmath, Martin. 2021b. General linguistics must be based on universals (or nonconventional aspects of language). *Theoretical Linguistics* 47(1–2). 1–31. (doi:10.1515/tl-2021-2002)
- Haspelmath, Martin. 2023a. Coexpression and synexpression patterns across languages: Comparative concepts and possible explanations. *Frontiers in Psychology* 14. (doi:https://doi.org/10.3389/fpsyg.2023.1236853)
- Haspelmath, Martin. 2023b. Defining the word. *WORD* 69(3). 283–297. (doi:10.1080/00437956.2023.2237272)
- Haspelmath, Martin. 2023c. On what a construction is. *Constructions* 15(1). (doi:10.24338/cons-539)
- Haspelmath, Martin. 2023d. Types of clitics in the world’s languages. *Linguistic Typology at the Crossroads* 3(2). 1–59. (doi:10.6092/issn.2785-0943/16057)
- Haspelmath, Martin. 2024a. Four kinds of lexical items: Words, lexemes, inventorial items, and mental items. *Lexique: Revue en Sciences du Langage* (34). 71–95. (doi:10.54563/lexique.1737)
- Haspelmath, Martin. 2024b. Inflection and derivation as traditional comparative concepts. *Linguistics* 62(1). 43–77. (doi:10.1515/ling-2022-0086)
- Haspelmath, Martin. 2025. Roots and root classes in comparative grammar (to appear).
- Haspelmath, Martin. 2026. Biverbal predicate constructions and related concepts: Serial verbs, converbs, auxiliaries, and grammaticalization (to appear).
- Healy, Phyllis M. 1960. *An Agta grammar*. Manila: Bureau of Printing.
- Hippisley, Andrew & Stump, Gregory. 2016. Introduction. In Hippisley, Andrew & Stump, Gregory (eds.), *The Cambridge handbook of morphology*, 1–24. Cambridge: Cambridge University Press.
- Hockett, Charles F. 1954. Two models of grammatical description. *WORD* 10(2–3). 210–234.
- Jackendoff, Ray & Audring, Jenny. 2020. *The texture of the lexicon: Relational Morphology and the Parallel Architecture*. Oxford: Oxford University Press.
- Killian, Don. 2015. *Topics in Uduk phonology and morphosyntax*. Helsinki: University of Helsinki. (<https://helda.helsinki.fi/bitstreams/c48a91ed-6b36-412d-8edf-de69bb7efb59/download>)
- Lieber, Rochelle. 1980. *On the organization of the lexicon*. Cambridge, MA: Massachusetts Institute of Technology. (PhD Thesis.) (<https://dspace.mit.edu/bitstream/handle/1721.1/15976/07044662-MIT.pdf?sequence=2>)
- Lieber, Rochelle. 1992. *Deconstructing morphology: Word formation in syntactic theory*. Chicago: University of Chicago Press.
- Luís, Ana R. 2023. Exploring divergent views on word structure: Challenging the concept of the morpheme. *Biblos* (9). 277–299. (doi:10.14195/0870-4112_3-9_13)
- Lyons, John. 1968. *Introduction to theoretical linguistics*. Cambridge: Cambridge University Press.
- Malchukov, Andrej. 2011. Interaction of verbal categories: Resolution of infelicitous grammeme combinations. *Linguistics* 49(1). 229–282. (doi:10.1515/ling.2011.006)
- Matthews, Peter H. 1972. *Inflectional morphology*. Cambridge: Cambridge University Press.
- Matthews, Peter H. 1991. *Morphology*. 2nd edition. Cambridge: Cambridge University Press.
- Mel’čuk, Igor A. 1979. Syntactic, or lexical, zero in natural language. *Berkeley Linguistics Society* 5. 224–260.
- Mel’čuk, Igor A. 2006. *Aspects of the theory of morphology*. Berlin: De Gruyter. (<https://doi.org/10.1515/9783110199864>)
- Meyerhoff, Miriam. 2013. *Bislama structure dataset*. In *APiCS Online (apics-online.info)*.
- Mithun, Marianne. 2014. Morphology: What’s in a word? In Genetti, Carol (ed.), *How languages work: An introduction to language and linguistics*, 71–99. Cambridge: Cambridge University Press.
- Müller, Gereon. 2004. On decomposing inflection class features: Syncretism in Russian noun inflection. In Müller, Gereon & Gunkel, Lutz & Zifonun, Gisela (eds.), *Explorations in nominal inflection*, 189–227. Berlin: Mouton de Gruyter.
- Nakanishi, Kimiko & Tomioka, Satoshi. 2004. Japanese plurals are exceptional. *Journal of East Asian Linguistics* 13(2). 113–140. (doi:10.1023/B:JEAL.0000019058.46668.c1)

- Okimāsis, Jean L. 2018. *Cree: Language of the Plains / nēhiyawēwin: paskwāwi-pīkiskwēwin*. Regina: University of Regina Press. (<https://hdl.handle.net/10294/8401>)
- Paul, Hermann. 1880. *Principien der Sprachgeschichte*. Halle: Niemeyer.
- Pesetsky, David. 2013. *Russian case morphology and the syntactic categories*. Cambridge, MA: MIT Press.
- Philippova, Nailya. 2021. *A grammar of Amri Karbi*. University of Helsinki Helsinki, Finland. (PhD Thesis.) (<https://helda.helsinki.fi/bitstreams/3c8326e5-c3bc-4bfb-9636-284466f4c2cc/download>)
- Popova, Gergana & Spencer, Andrew. 2013. Relatedness in periphrasis. In Chumakina, Marina & Corbett, Greville (eds.), *Periphrasis: The role of syntax and morphology in paradigms*. Oxford: British Academy. (doi:10.5871/bacad/9780197265253.001.0001)
- Roberts, John R. 1987. *Amele* (Croom Helm Descriptive Grammars). London: Croom Helm.
- Ryding, Karin C. 2005. *A reference grammar of Modern Standard Arabic*. Cambridge: Cambridge University Press.
- Selkirk, Elisabeth O. 1982. *The syntax of words*. Cambridge, MA: The MIT Press.
- Siddiqi, Daniel. 2019. Distributed Morphology. In Audring, Jenny & Masini, Francesca (eds.), *The Oxford handbook of morphological theory*, 143–165. Oxford: Oxford University Press. (<https://doi.org/10.1093/oxfordhb/9780199668984.001.0001>)
- Spencer, Andrew. 2019. Theoretical approaches to morphology. In Aarts, Bas & Bowie, Jill & Popova, Gergana (eds.), *The Oxford Handbook of English Grammar*, 56–82. Oxford: Oxford University Press.
- Stewart, Thomas W. 2016. *Contemporary morphological theories: A user's guide*. Edinburgh: Edinburgh University Press.
- Stump, Gregory T. 2001. *Inflectional morphology: A theory of paradigm structure*. Cambridge: Cambridge University Press.
- Stump, Gregory T. 2016. Paradigms at the interface of a lexeme's syntax and semantics with its inflectional morphology. In Siddiqi, Daniel & Harley, Heidi (eds.), *Morphological metatheory*, 27–58. Amsterdam: Benjamins.
- Stump, Gregory T. 2019. Theoretical issues in inflection. In Audring, Jenny & Masini, Francesca (eds.), *The Oxford handbook of morphological theory*, 56–82. Oxford: Oxford University Press. (doi:10.1093/oxfordhb/9780199668984.013.4)
- Trommer, Jochen. 2012. Ø-exponence. In Trommer, Jochen (ed.), *The morphology and phonology of exponence*. Oxford: Oxford University Press.
- van der Auwera, Johan & Gast, Volker. 2011. Categories and prototypes. In Song, Jae Jung (ed.), *The Oxford handbook of linguistic typology*, 166–189. Oxford: Oxford University Press.
- Vicente, Agustín & Falkum, Ingrid L. 2017. Polysemy. *Oxford Research Encyclopedia of Linguistics*. (doi:10.1093/acrefore/9780199384655.013.325)
- von Prince, Kilu & Krajinović, Ana & Krifka, Manfred. 2022. Irrealis is real. *Language* 98(2). 221–249. (doi:<https://dx.doi.org/10.1353/lan.2022.0009>)
- Williams, Edwin. 1981. Argument structure and morphology. *The Linguistic Review* 1(1). 81–114. (doi:10.1515/tlir.1981.1.1.81)
- Wurzel, Wolfgang Ullrich. 1984. *Flexionsmorphologie und Natürlichkeit: Ein Beitrag zur morphologischen Theoriebildung*. Berlin: Akademie-Verlag.
- Wurzel, Wolfgang Ullrich. 1989. *Inflectional morphology and naturalness*. Dordrecht: Kluwer.
- Yule, George. 2010. *The study of language (4th edition)*. Cambridge: Cambridge University Press.
- Zhang, Qiao. 1998. Fuzziness - vagueness - generality - ambiguity. *Journal of Pragmatics* 29(1). 13–31. (doi:10.1016/S0378-2166(97)00014-3)
- Zwicky, Arnold M. 1985. How to describe inflection. *Annual Meeting of the Berkeley Linguistics Society* 11. 372–386.